Supplementary Information

The following information is associated with Network Rail Standard: **NR/L2/CIV/177** Issue 1

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Word Forms

- **NR/L2/CIV/177/F001**

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Ref:	NR/L2/CIV/177
Issue:	1
Date:	03 June 2017
Compliance date:	02 September 2017

Level 2

Business process

Monitoring track over or adjacent to building and civil engineering works

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User information

This Network Rail standard contains colour-coding according to the following Red–Amber–Green classification.

Red requirements – *no variations permitted*

- Red requirements are to be complied with and achieved at all times.
- Red requirements are presented in a red box.
- Red requirements are monitored for compliance.
- Non-compliances will be investigated and corrective actions enforced.

Amber requirements – *variations permitted subject to approved risk analysis and mitigation*

- Amber requirements are to be complied with unless an approved variation is in place.
- Amber requirements are presented with an amber sidebar.
- Amber requirements are monitored for compliance.
- Variations can only be approved through the national variations process.
- Non-approved variations will be investigated and corrective actions enforced.

Green guidance – *to be used unless alternative solutions are followed*

- Guidance should be followed unless an alternative solution produces a better result.
- Guidance is presented with a dotted green sidebar.
- Guidance is not monitored for compliance.
- Alternative solutions should be documented to demonstrate effective control.

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Compliance

This Network Rail standard/control document is mandatory and shall be complied with by Network Rail Limited and its contractors if applicable from 2nd September 2017.

Where it is considered not reasonably practicable[1] to comply with the requirements in this standard/control document, permission to comply with a specified alternative should be sought in accordance with the Network Rail standards and controls process, or with the Railway Group Standards Code if applicable.

If this standard/control document contains requirements that are designed to demonstrate compliance with legislation they shall be complied with irrespective of a project's GRIP stage. In all other circumstances, projects that have formally completed GRIP Stage 3 (Option Selection) may continue to comply with any relevant Network Rail standards/control documents that were current when GRIP Stage 3 was completed.

NOTE 1: *Legislation includes Technical Specifications for Interoperability (TSIs).*

NOTE 2: *The relationship of this standard/control document with legislation and/or external standards is described in the purpose of this standard.*

Disclaimer

In issuing this standard/control document for its stated purpose, Network Rail Limited makes no warranties, expressed or implied, that compliance with all or any standards/control documents it issues is sufficient on its own to provide safety or compliance with legislation. Users are reminded of their own duties under legislation.

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[1] *This can include gross proportionate project costs with the agreement of the Network Rail Assurance Panel (NRAP).*

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Issue record

Issue	Date	Comments
1	June 2017	First issue.

Reference documentation

NR/L2/CIV/003	*Engineering Assurance of Building & Civil Engineering Works*
NR/L2/TRK/001/mod11	*Track geometry - Inspections and minimum actions*
NR/L2/TRK/001/mod12	*Track Geometry - Maintenance design requirements*
NR/L2/TRK/001/mod14	*Managing track in hot weather*
NR/L2/INF/02202	*Records management of health and safety files*
NR/L2/MTC/088	*Arrangements for maintenance of new and changed assets*
NR/L2/CIV/177/F001	*Track Monitoring Plan (template)*
NR/L1/INI/PM/GRIP/100	*Governance of Railway Investment Projects (GRIP) – Policy*
RIS-1530-PLT	*Rail Industry Standard for Technical Requirements for On-Track Plant and Their Associated Equipment and Trolleys*

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Table 1 – Minimum action to be taken following detection of geometry faults on constructions sites 12

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1 Purpose

This business process defines the requirements for monitoring of the track over or adjacent to building and civil engineering works to maintain:

a) safe operation of trains; and / or

b) the safe movement and control of people to and from the trains.

It contains specific requirements to manage track safety during building and civil engineering works.

2 Scope

This business process applies to:

a) the construction and commissioning of building and civil engineering works on or adjacent to Network Rail infrastructure;

b) anyone who is responsible for managing these phases of the work; and

c) Network Rail and outside party projects.

This business process does not apply to non-operational lines.

The requirements of monitoring stated in this business process may be adopted for other infrastructure where impacted by the building or civil engineering works.

In following this procedure, reference should be made to the GRIP stages defined in NR/L1/INI/PM/GRIP/100.

3 Definitions

Approval in Principle

Technical submission as required by NR/L2/CIV/003.

Asset Management Plan

The document that details how assets are maintained during the works and when introduced into operational use.

Critical Rail Temperature

The rail temperature to which track is allowed to rise before specific measures to protect traffic are required.

Outside Party

An organisation, other than Network Rail, that is an infrastructure owner or developer, or a user or occupier of Network Rail's infrastructure. The term includes highway authorities, passenger transport executives, public or private developers, public utilities and train operating companies.

For the implementation of this standard, the term includes for Third Party schemes.

Permissible Speed (or Enhanced Permissible Speed, EPS)

The maximum speed published in the Sectional Appendix at which trains are permitted to travel over a section of track.

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Road Rail Vehicle

A road-rail vehicle (RRV) is one that can travel on the ground and also travel on rail by virtue of a rail wheel guidance system under its own power system. Such machines are not allowed to operate, work or travel on rail outside possessions.

Statement of Design Intent

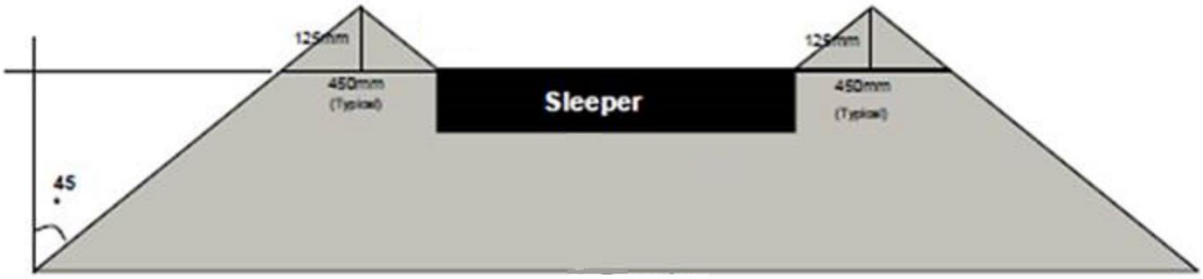
The basis of the Design of the Works (or a specific part of them), defined within NR/L2/CIV/003/F002.

Track Recording Vehicle

Track borne vehicle with the ability to record track geometry and identify other track faults without the need for manual inspections.

Track Support Zone

The area bounded within a 45 degree plane extending below and beyond the top of the ballast shoulder. The diagram below shows the Track Support Zone, where the ballast shoulder is assumed to have a width of 450mm and a depth of 125mm from the top of the sleeper. These dimensions may vary on location.



Track System

This is used as a collective term for the parts that make up the permanent way.

NOTE This term includes rail, sleepers, fastening, ballast and track bed.

Zone of Influence

The volume of ground adjacent to building and civil engineering works, which might be affected during construction and their operational life, e.g. through structural instability or collapse.

NOTE The zone of influence will need to be determined for each project as it will vary depending upon various factors including ground condition and type of building or civil engineering works.

4 Track Monitoring Requirements

Track geometry shall be monitored if:

a) building or civil engineering works are either under or adjacent to the track support zone; or

b) a ground movement assessment identifies the potential for track movement.

An assessment of the potential ground movement, due to the temporary and permanent construction for building and civil engineering works, shall be undertaken.

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The results of the assessment shall be reviewed and incorporated in the Statement of Design Intent stage of the project.

The assessment shall take into account:

a) zone of influence interaction with track support zone;

b) ground movement caused by excavation;

c) ground movement caused by soil displacement;

d) ground movement caused by loading or unloading of the soil;

e) ground movement caused by de-watering;

f) vibration.

NOTE 1 The requirements of monitoring stated in this business process may be adopted for other infrastructure within the zone of influence of the building and civil engineering works.

The Principal Contractor shall analyse the track monitoring against predicted ground movement. They shall then set agreed project specific trigger levels and actions within the Asset Management Plan (AMP).

Interventions and actions required shall be:

a) agreed as part of an escalation and contingency plan prior to the start of the works; and

b) recorded in the Track Monitoring Plan (TMP).

Table 1 shall be used as the minimum actions to be taken following detection of geometry faults during building and civil engineering works. The Designer shall carry out a risk assessment as part of the design to advise whether the minimum trigger levels need to be enhanced further.

NOTE 2 Response times for geometry faults created as a result of construction works require enhanced rectification timescales. This is because these faults are likely to deteriorate more rapidly than those caused by normal track deterioration.

The actual track limiting values shall be determined in agreement with the Track Maintenance Engineer (TME).

NOTE 3 The TME can advise on any local track geometry constraints which might affect the limiting values within table 1.

If Road Rail Vehicles (RRV) will be used during the building or civil engineering works, RIS-1530-PLT shall be used to verify the track limiting values identified in table 1. If more onerous track limiting values are determined through use of RIS-1530-PLT they shall replace those identified in table 1.

NOTE 4 Track limiting values may be more onerous than those listed in table 1 due to the shorter wheel base of RRVs.

The Project Manager shall:

a) make arrangements for the measurements to be reviewed; and

b) consult the Infrastructure Maintenance Delivery Manager's (IMDM) representative on the appropriate course of action as an intervention level is approached.

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NOTE 5 This action might include a change in the monitoring regime.

If dynamic measurements are obtained from a Track Recording Vehicle (TRV), the IMDM's representative and the Project Manager shall arrange for these measurements to be merged with the data obtained from the site monitoring regime.

NOTE 6 This is to confirm the adequacy of the track or, conversely, to identify any remedial action that is required and the timescale for such action.

A copy of the recorded measurements and calculated movements against the baseline, along with confirmation of any action taken, shall be submitted to the IMDM representative(s), as defined in the AMP.

5 Track Monitoring plan

The Project Manager shall produce an AMP in accordance with NR/L2/MTC/088.

The AMP shall:

a) be agreed with the TME;

b) include a TMP.

NOTE A template for a TMP is set out in NR/L2/CIV/177/F001.

The TMP shall confirm:

a) the roles and responsibilities for monitoring, recording, interpreting and reporting the data from the monitoring works are defined;

b) the person responsible on site for the day to day management of the data from the monitoring works is confirmed;

c) communication links between the project and the maintenance management organisation are defined and demonstrated to work;

d) the roles and responsibilities for rectifying track faults are defined;

e) the names of individuals with post holder responsibilities for all the parties involved,

f) full details of the monitoring regime are recorded;

g) the location, extent and duration of the monitoring is fully defined; and

h) affected Network Rail assets (other than the track) are identified and managed appropriately by the project.

The TMP shall be reviewed against Approved for Construction drawings to confirm compatibility prior to execution of the works.

The form and method of reporting the monitoring results shall be agreed and documented before work commences on the site.

Changes in the gauge, geometry and position of the track shall be determined from:

a) optical measurements; or

b) through other monitoring methods if their accuracy can be reliably demonstrated.

To minimise access to the track, targets may be installed on the rails or sleepers.

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Measurements shall be related to a stable benchmark(s) located outside the zone of influence of the site works, determined by the project. Measurement shall be compared against baseline readings taken prior to the start of the works.

6 Baseline condition requirements

Prior to any work commencing on site, arrangements shall be made with the TME to assess and record the condition of the track.

NOTE 1 This might include the presence of closed joints or other track components.

NOTE 2 For Outside Party projects, the organisation responsible for the construction of the works onsite may arrange for a condition and dilapidation survey to be carried out (with the TME's approval).

The Project Team shall review the condition of the track and check this complies with the maintenance limits for gauge, geometry and position defined in NR/L2/TRK/001/mod11and NR/L2/TRK/001/mod12.

If it does not, the Project Team shall work with the TME to make arrangements to achieve compliance prior to commencement of the works.

A final set of baseline readings shall be taken no more than 7 days prior to any work commencing on site.

Existing data from a TRV may be used to define track geometry prior to commencement of any site work.

7 Specific monitoring requirements

Where track monitoring is required, it shall be supplemented with subsurface monitoring within the track support zone if there is the potential for the permanent way to bridge a void, masking the severity of a potential problem, formed by the construction works.

NOTE 1 One potential method to achieve this is to install sub-surface monitoring rods extending throughout the anticipated area of influence.

NOTE 2 Fluctuating readings might be an indication that voids are forming under the track. Other causes of fluctuation might be monitoring tolerance, climatic effects on the monitoring equipment and vibration effects from the operational railway or proximate construction activities.

Where practicable, the ground surface below the ballast should be made visible to allow for routine inspections during the course of the work.

Both surface and subsurface measurements shall be taken at 3 metre intervals between points lying 30 metres outside either end of the area where the track might be affected by the works.

NOTE 3 For example, where the zone of influence is predicted to be 40m either side of the construction works, monitoring shall extend 70m.

The frequency of measurements of the track shall be based on a risk assessment of the site conditions. The risk assessment shall take into account:

a) any pre-existing movement trends;

b) the line speed;

c) the staging of the works;

d) prevailing weather conditions;

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e) the type of works.

The TME shall agree the frequency of measurements.

As a minimum, monitoring readings shall be analysed daily throughout the duration of the works.

8 Critical rail temperature

Any building or civil engineering works conducted adjacent to, within, or underneath the track support zone are considered as a disturbance. The disturbance might also create deficiencies in the track system.

If the works have the potential to cause a disturbance or deficiency, the critical rail temperature shall be calculated, monitored and managed in accordance with NR/L2/TRK/001/mod14.

9 Monitoring following completion of the works

Monitoring shall continue following completion of the works. Measurements shall be taken:

a) daily in the week following the end of the works;

b) twice a week in the second and third weeks (at least 3 days apart); and

c) a final set of measurements in the fourth week (the date/time to be agreed).

If movement is apparent by the end of the third week, monitoring shall continue until track movement has ceased.

10 Records

The type of records required for each project will vary depending on the building and civil engineering works being carried out.

The installer shall agree with either the Network Rail Project Manager or Designated Project Engineer the type of records to be kept, their format and timescale for submission.

The Approval in Principle shall list the records to be kept to support the procedures outlined in the relevant Network Rail standards.

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11 Intervention limits and actions

If triggers levels are being reached, consider reviewing the monitoring frequency to aid identification of trends at an earlier stage to prevent future triggers.

Fault ref.	Measure	Limiting Value	Action
1	Cross level [1] (cant / super elevation)	Difference in consecutive cants measured 3 metres apart is greater than 10 mm (equivalent to > 1 in 300 twist)	Note. No immediate action.
2	Cross level [1] (cant / super elevation)	Difference in consecutive cants is greater than 15 mm (equivalent to a greater than a 1 in 200 twist)	Report to Infrastructure Control Centre (ICC) [2]. Rectify fault within 7 days.
3	Cross level [1] (cant / super elevation)	Difference in consecutive cants is greater than 24 mm (equivalent to a greater than a 1 in 125 twist)	Report to ICC [2]. Stop work [3]. Rectify fault within 24 hours.
4	Cross level [1] (cant / super elevation)	Difference in consecutive cants is greater than 33 mm (equivalent to a greater than a 1 in 91 twist)	BLOCK THE LINE (contact signaller direct) Report to ICC [2]. Stop work [3]. Rectify fault before opening the line. Increase monitoring frequency until settlement/heave has stabilised.
5	Cant (super elevation). On non-electrified track	Change in cant from original datum position of ± 20 mm or greater	Report to ICC [2]. Stop work until track has been inspected by SM[T] (Section Manager [Track]). Correct fault as directed by SM[T]. Review method of working.
6	Cant (super elevation). On OLE electrified track	Change in cant from original datum position of ± 15 mm or greater	
7	Displacement from original datum position [4]. Non- Enhanced Permissible Speed (EPS) track	25 mm or more in any direction	
8	Displacement from original datum position [4]. Enhanced Permissible Speed (EPS) track	10 mm or more in any direction	

Notes:

1 Measured across the two rails of the same track, measured 3 metres apart.

2 ICC shall inform SM [T] (Section Manager [Track]).

3 Review method of working before continuing.

4 These dimensional tolerances may be reduced if the work is taking place in a tight clearance location.

Table 1 – Minimum action to be taken following detection of geometry faults on constructions sites

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Standard and control document briefing note

Ref: NR/L2/CIV/177	**Issue:** 1
Title: Monitoring track over or adjacent to building and civil engineering works	
Publication date: 3rd June 2017	**Compliance Date:** 2nd September 2017
Standard/Control Document Owner: Head of Tunnels	
Non-compliance rep (Approver of TRACKER applications): Robert Anderson, Senior Engineer	
Technical lead/contact for briefings: Robert Eggleton / Luigi Rocco	**Tel:** 07730 352509

Purpose:
This business process defines the requirements for monitoring of the track over or adjacent to building and civil engineering works to maintain:

a) safe operation of trains; and / or

b) the safe movement and control of people to and from the trains.

It contains specific requirements to manage track safety during building and civil engineering works.

Scope:
This business process applies to:

a) the construction and commissioning of building and civil engineering works on or adjacent to Network Rail infrastructure;

b) anyone who is responsible for managing these phases of the work; and

c) Network Rail and outside party projects.

This business process does not apply to non-operational lines.

The requirements of monitoring stated in this business process may be adopted for other infrastructure where impacted by the building or civil engineering works.

In following this procedure, reference should be made to the GRIP stages defined in NR/L1/INI/PM/GRIP/100.

What's new/ what's changed:

This is a new standard/control document.

This standard has been produced to replace NR/BS/LI/045; the majority of the content is the same. The scope and purpose of this document are detailed above.

Additional content have been included surrounding Critical Rail Temperature, Asset Management Plans, Monitoring of the ground and void detection. The intervention limits have been modified from the table previously in NR/BS/LI/045.

***NOTE:** It is the duty of those briefed or notified, to read through this document and familiarise themselves with its content.*

Section/clause	Amended/ deleted/ new	Summary of changes
All	New	Standard has been produced incorporating modified clauses from NR/BS/LI/045, a number of clauses have been modified to incorporate learning since its use and new sections added to address recommendations.
7	Amended	The standard now required you to make the ground visible for inspection wherever practicable during the works.
8	New	Critical Rail Temperature included as a monitoring requirement
11	Deleted	There was a table of intervention limits in LI/045, this has been reviewed and an updated table has been included. It is noted that these are the minimum limits and that the values should be reviewed for each project.
Form 1	New	Template produced for a Track Monitoring Plan

***NOTE 1:** For revised standards/control documents you may summarise general changes to content and list significant new/revised content.*
***NOTE 2:** For new standards/control documents you may summarise the content of the entire document rather than list each clause separately.*

Reasons for change:
Include benefits, risks mitigated, recommendations closed out etc.

- Stoke Lane Recommendation 4
- Bradwell Abbey Recommendation 6
- Replacement of Letter of Instruction NR/BS/LI/045.

Affected documents:

Reference	*Impact*
NR/L2/CIV/177 ISSUE 1	New
NR/L2/CIV/177/F001 ISSUE 1	New
NR/BS/LI/045	Withdrawn

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Briefing requirements:

Technical briefings are given to those who have specific responsibilities within this standard/control document.

Awareness briefings are given to those who might be affected by the content but have no specific responsibilities within the standard/control document.

Details of the briefing arrangements are included in the associated briefing programme.

Briefing *(A-Awareness/ T-Technical)*	**Post**	**Function**	**Responsible for cascade briefing? Y/N**
T	Asset Protection Project Manager	Network Operations	N
T	Asset Protection Engineer	Network Operations	N
T	Senior Asset Protection Engineer	Network Operations	Y
T	Programme Engineering Manager	Infrastructure Projects [Buildings and Civils]	Y
T	Senior Project Engineer [Delivery Services]	Route Services	Y
T	Programme Manager [Works Delivery] [Buildings and Civils]	Network Operations	Y
T	Project Manager [Works Delivery] [Buildings and Civils]	Network Operations	N
A	Head of Track	STE	Y
A	Head of Maintenance	STE	Y
A	Head of Signalling	STE	Y
A	Head of Power Distribution HV/LV	STE	Y
A	Head of Structures	STE	Y
A	Head of Buildings	STE	Y
A	Head of Geotechnics	STE	Y
A	Head of Drainage	STE	Y

NOTE: *Contractors are responsible for arranging and undertaking their own Technical and Awareness Briefings in accordance with their own processes and procedures.*

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